Medical Image Analysis 101 (2025) 103458

Contents lists available at ScienceDirect

Medical Image Analysis

journal homepage: www.elsevier.com/locate/media

VSNet: Vessel Structure-aware Network for hepatic and portal vein segmentation

Jichen Xu [a,1], Anqi Dong [b,1], Yang Yang [c], Shuo Jin [d], Jianping Zeng [d], Zhengqing Xu [e,f], Wei Jiang [g], Liang Zhang [h], Jiahong Dong [d], Bo Wang [i,e,f,*]

[a] *National Key Laboratory of Human–Machine Hybrid Augmented Intelligence, National Engineering Research Center for Visual Information and Applications, and Institute of Artificial Intelligence and Robotics, Xi'an Jiaotong University, Xi'an, China*
[b] *Division of Decision and Control Systems and Department of Mathematics, KTH Royal Institute of Technology, SE-100 44 Stockholm, Sweden*
[c] *School of Automation, Northwestern Polytechnical University, Xi'an, China*
[d] *Beijing Tsinghua Changgung Hospital, School of Clinical Medicine, Tsinghua University, Beijing, China*
[e] *Jingzhen Medical Technology Ltd., China*
[f] *Matrix Medical Technology Ltd., China*
[g] *Research Center of Artificial Intelligence of Shangluo, Shangluo University, Shangluo, China*
[h] *School of Computer Science and Technology, Xidian University, Xi'an, China*
[i] *Institute of Medical Equipment Science and Engineering, Huazhong University of Science and Technology, Wuhan, China*

A R T I C L E I N F O

MSC:
68T07
68Uxx
92C50

Keywords:
Liver vessel segmentation
Multi-task learning model
Convolution neural network

A B S T R A C T

Identifying and segmenting hepatic and portal veins (two predominant vascular systems in the liver, from CT scans) play a crucial role for clinicians in preoperative planning for treatment strategies. However, existing segmentation models often struggle to capture fine details of minor veins. In this article, we introduce Vessel Structure-aware Network (VSNet), a multi-task learning model with vessel-growing decoder, to address the challenge. VSNet excels at accurate segmentation by capturing the topological features of both minor veins while preserving correct connectivity from minor vessels to trucks. We also build and publish the largest dataset (303 cases) for hepatic and portal vessel segmentation. Through comprehensive experiments, we demonstrate that VSNet achieves the best Dice for hepatic vein of 0.824 and portal vein of 0.807 on our proposed dataset, significantly outperforming other popular segmentation models. The source code and dataset are publicly available at https://github.com/XXYZB/VSNet.

1. Introduction

Increasing attention has been drawn to *hepatocellular carcinomas* (HCC) — one of the most prevalent liver cancers with a top death rate among all other cancers (Asrani et al., 2019). Various treatments are proposed for HCC, e.g., surgical resection, thermal ablation, and interventional therapy (Chen et al., 2020a). The success of such therapeutic approaches relies on meticulous and correct preoperative planning, that requires an accurate extraction of the anatomy of hepatic vessels from patients' computed tomography (CT) images. Two sets of veins are considered in clinic for preoperative planning: *hepatic veins* (HV) that are responsible for draining venous blood from the liver, and *portal veins* (PV) conveys blood to the liver. Despite their interconnected nature through small venous sinuses, these connections are not detectable in CT scans. Thus, the structure of two veins characterized by CT scans, ideally forms two trees that are disconnected from each other. The precise identification and segmentation of HV and PV are key factors in preoperative planning since the treatment strategies may differ for them. For example, PV needs to be occluded during anatomic resection of liver, whereas it is usually not required on HV (van Gulik et al., 2007).

Currently, radiologists achieve liver vessel segmentation manually, and hence laborious and time-consuming. Moreover, the spatial distances between the alternative distal vessels are relatively smaller than those between peripheral vessels and trunks in the same category, and the intensities of the two bunches have almost no difference on the CT images, thereby easily causing misclassification on vessels from different trunks (Schindera et al., 2006). The misclassification can lead to errors in liver segment classification, thus affecting the prediction of the resection plane in hepatectomy (Heriot and Karanjia, 2002). To this end, we develop a segmentation model that automatically extracts

* Corresponding author at: Institute of Medical Equipment Science and Engineering, Huazhong University of Science and Technology, Wuhan, China.
E-mail address: bow@mail.hust.edu.cn (B. Wang).
[1] Jichen Xu and Anqi Dong contributed equally to this work.

https://doi.org/10.1016/j.media.2025.103458
Received 14 January 2024; Received in revised form 13 November 2024; Accepted 7 January 2025
Available online 16 January 2025

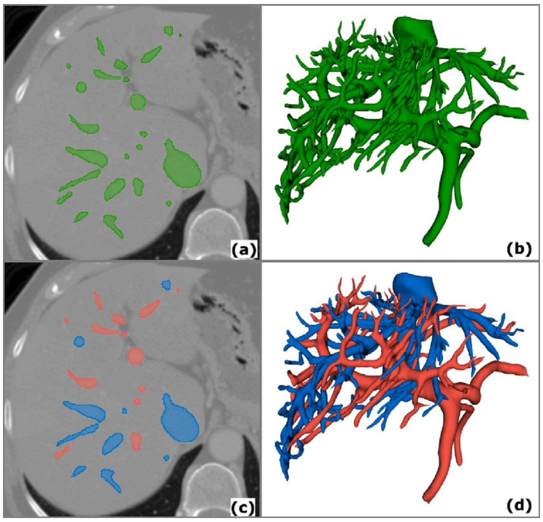

Fig. 1. Comparison of a liver vessel and separate HV/PV. Fig. (a) and (b) are the masks of liver vessels on a slice and 3D space. Fig. (c) and (d) show the separately annotated masks of HV and PV on a slice and 3D space.

tree-structure HP and PV from CT scans. Different from joint liver vessel segmentation, our framework aims to identify and categorize the mask HP and PV separately according to their connection to the trunk, see also Fig. 1.

Deep learning-based segmentation models for HV and PV have been proposed in the recent decade. For example, the frameworks based on 3D-Unet (Yu et al., 2019) and V-Net (Ivashchenko et al., 2021) are designed for HV/PV segmentation on CT images and MR images, respectively. However, they neglected to tackle the easily generated misclassification of peripheral vessels. *Keshwani et al.* proposed a multi-task approach to solving the problem by employing auxiliary tasks to estimate connectivity between voxels on vascular centerlines, which helps to reconstruct vascular tree structures (Keshwani et al., 2020). Such an approach still cannot effectively solve the issue because centerlines in this method are constructed by sparse points, and thereby, are likely to omit the corresponding parts of the peripheral vessels for HV and PV.

We introduce a multi-task HV/PV segmentation model, namely, the Vessel Structure-aware Network (VSNet), to address the concerns mentioned above. Additionally, VSNet incorporates two auxiliary tasks aiming at enhancing the model's performance for vessel segmentation. One is the dense centerline regression, that depicts the complete skeletons of HV and PV with the same saliency for minor and main vessels. The second is to segment the edges of HV and PV. It is exploited to preserve connectivities between small and superior vessels since accurately segmented boundaries can enclose a tree-like venous structure into a connected area. The three tasks share weights before the last layer of the decoder for better knowledge interaction.

We also design a vessel-growing decoder that simulates the progressive extension of minor vessels from the main vessels to establish correct connections from minor vessels to superior vessels. It helps to avoid misclassifying small vessels because they are usually caused by the wrong associations between small vessels and superior vessels generated in the decoder, which can be transmitted with decoding. Specifically, we introduce a *Global Dual Transformer* (GDT) to obtain the spatial dependency of main HV and PV in high-level features. The spatial position relationship is passed to the upsampled features through a proposed *Dependency Transmitting* (DT) block. Combining with the features from the corresponding layer in the encoder, features of secondary vessels associated with the superior vessel are grown. The deep supervision mechanism is applied to supervise the growth.

In addition, the eminent performance of deep learning algorithms depends on high-quality annotated datasets. However, public vessel segmentation datasets with high-quality HV and PV annotations are scarce. To resolve this problem, we re-annotate all the CT volumes with masks of HV and PV from the training set of liver vessel segmentation tasks in the Medical Segmentation Decathlon (MSD) challenge and make the annotations available to the public. The dataset includes 303 CT volumes containing the liver region, with slice thickness ranging from 0.8 mm to 8 mm. The original labels do not indicate HV and PV separately and have many omissions and errors (Xu et al., 2021; Su et al., 2021; Svobodova et al., 2022; Liu et al., 2023). To the best of our knowledge, the proposed dataset is the largest public dataset for HV/PV segmentation.

The contributions of this article are summarized as follows:

- A multi-task-based model (VSNet) with a vessel-growing decoder and two auxiliary tasks is proposed for hepatic and portal vein segmentation.
- We re-annotate the hepatic vessel segmentation dataset in the MSD challenge and make it public. The hepatic vein and portal vein are separately annotated in the revised dataset.
- The proposed model is evaluated on the 3DIRCADb dataset and the revised MSD dataset. The proposed model (VSNet) outperforms other state-of-the-art methods.

In the remainder of this article, we first briefly review multi-task learning and existing vessel segmentation methods in Section 2. The structure of the proposed VSNet and details of the proposed modules are introduced in Section 3. We demonstrate the settings of experiments, evaluation metrics and results in Section 4. Finally, we conclude in Section 6 with future research directions.

2. Related works

2.1. Multi-task learning

Multi-task learning involves simultaneously addressing at least two tasks with shared parameters and has proven effective in the field of computer vision (Chen et al., 2020b; Ramamonjisoa and Lepetit, 2019). Such an approach aims to enhance the performance of the main task by leveraging inductive knowledge obtained from auxiliary tasks. In comparison to single-task learning, the associated tasks in multi-task learning complement each other, thereby improving the models' overall generality. Additionally, this approach also increases inference speed for multiple tasks and reduces memory occupation due to shared layers. Multi-task learning has been increasingly attempted in medical image analysis across various human anatomical regions, which is attributed to its efficiency in exploiting the inherent relationships among tasks and different types of medical image annotations (Zhao et al., 2022). Applications can be found in brain (Lian et al., 2021), eye (Chelaramani et al., 2021), chest (Park et al., 2022) and, abdomen (Yao et al., 2021), and more over, works combine the segmentation of multiple regions-of-interest with image regression (Kuang et al., 2021), classification (Liu et al., 2020), registration (Chen et al., 2019), and reconstruction (Estienne et al., 2019).

Multi-task-based methods have also been proposed for segmentation of organ images, for instance, intestines and cardiovascular parts. Models for intestinal tumor segmentation often adopt classification of lesions as the auxiliary task (Zhang et al., 2021b; Tang et al., 2023), that is unhelpful for segmenting vascular structures. When segmenting coronary artery, additional annotations like calcium score (Zhang et al., 2021c) or landmarks in atria and ventricle (Zhang et al., 2021a) may be needed. Zhang et al. (2023) proposed a method that employs centerline segmentation as the auxiliary task, which may cause an imbalance of foreground and background. Liu et al. (2022) utilize auxiliary tasks based on distance map and edge segmentation to avoid drawbacks mentioned above, but may be only efficient for organs with simple

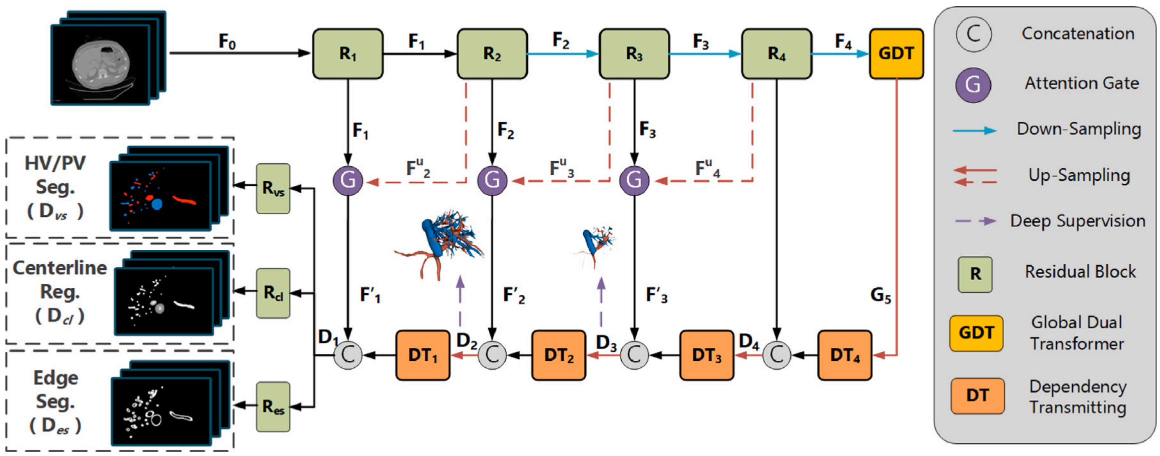

Fig. 2. The overall structure of the proposed model with CT images as input. At the encoder stage, four consecutive Residual blocks are adopted to process the features. The attention mechanism is used between the output features of two adjacent layers to assign spatial attention to the low-level features. Then, the features enter the GDT module to obtain global dual dependencies. Afterward, the features will pass through four series-connected DT blocks. Finally, the output of the last DT block enters three parallel residual blocks to obtain predictions for three different tasks.

topological structures. In this work, we proposed a regression task with a newly designed distance map and a novel edge segmentation task as auxiliaries, tackling to solve the difficulties of HV/PV segmentation.

2.2. Liver vessel segmentation

Various approaches for automatic liver vessel segmentation have been proposed and studied in the recent few decades. Traditional methods involve image-processing techniques such as region growing (Hao et al., 2000), contour-based level-set methods (Isgum et al., 2009), adaptive thresholding (Guo et al., 2015), and image filters (Qian, 2017). Specifically, for HV and PV segmentation, tracing-based approaches are employed to delineate the topology structure of both veins (Goceri, 2016; Lebre et al., 2019). However, these strategies require starting points on volumes, such as end node (Kang et al., 2014) of vessel centerlines or leaf nodes along the height ridge of vessels (Zeng et al., 2017). The performance of these methods highly depends on appropriate initialization, and human intervention is regularly requested. Deep learning approaches have recently been prevalent due to their efficiency in extracting image features, less dependence on manual initialization, and better robustness. *Convolution Neural Networks* (CNN) and *Transformer* are extensively used deep learning techniques in hepatic vessel segmentation. *Kitrungrotsakul et al.* introduced three deep convolutional networks to segment hepatic vasculature in CT images (Kitrungrotsakul et al., 2017). *Wu et al.* utilized Transformer to segment hepatic vessels on MR images. Many works have demonstrated that paying attention to vessel connections can help promote vessel segmentation efficiency. For instance, *Li et al.* used the multi-directional Sobel operator (Kittler, 1983) to extract connections in different directions (Li et al., 2022). *Yang et al.* explored the connectivity between pixels and adjacent pixels in different directions (Yang and Farsiu, 2023). Besides, the weighted loss function for solving the imbalance between vessels and background (Huang et al., 2018), attention mechanism (Yan et al., 2020), multi-scale learning (Hao et al., 2022), multi-scale feature fusion (Yan et al., 2020; Sun et al., 2023; Dong et al., 2023), and Laplacian salience filter (Gao et al., 2023) were introduced to improve the model's ability to extract detailed features. However, almost none of these methods can satisfactorily solve the misclassification of HV and PV peripheral blood vessels.

3. Methodology

3.1. Overall structure

The proposed multi-task learning model, whose main task lies on predicting masks of HV and PV based on CT volumes, follows an encoder–decoder structure (see Fig. 2). Despite the main task, the two auxiliary tasks are (1) regressing the distances from pixels within the vessel to the centerline, and (2) segmenting the boundaries of vessels to alleviate the misclassification of peripheral veins of HV and PV. In the specific encoder–decoder architecture, the encoder employs CNN-based residual blocks, while the decoder decodes the secondary vessel features associated with the superior vessels and passes them to the next level. As shown in Fig. 2, the Global Dual Transformer (GDT) block is utilized at the bridge to acquire the space-wise and channel-wise global dependency to describe the spatial relationship of superior HV and PV at the highest semantic level. Dependency Transmitting (DT) blocks are employed at the decoder to deliver the dependency from high-level to low-level features. The associated secondary vessels are thus reconstructed by skip connection of the outputs from the Dependency Transmitting block, and the features at the corresponding level from the encoder.

3.2. Feature extraction

We use four residual blocks at the encoder to extract low-level features. Denote F_i as the output of ith layer encoder ($i = 1, 2, 3, 4$). The procedure now reads

$$F_i = \mathrm{R}_i(F_{i-1}), \quad i = 1, 2, 3, 4 \tag{1}$$

where $\mathrm{R}_i(\cdot)$ denotes the ith residual block and F_0 is the original image. Max pooling operation is conducted from second to fourth blocks for downsampling. Spatial attention is accomplished by the attention gate on features between adjacent levels to amplify local texture information. Denote F_i' as the weighted features from the attention gate at the ith layer. The attention operation can be seen as

$$F_i' = F_i \otimes \sigma\left(\mathrm{Conv}\left(F_{i+1}^u \oplus F_i\right)\right), \quad i = 1, 2, 3, \tag{2}$$

where $\otimes$ and $\oplus$ are element-wise multiplication and element-wise addition, respectively. Here, F_i^u is the upsampled F_i obtained by deconvolutions, $\mathrm{Conv}(\cdot)$ is convolution operations, which squeeze the channels of added features into one, and σ is the sigmoid activate function.

The features are fed to the vessel-growing decoder right after, which consists of the GDT and DT blocks (see Section 3.3 for more detail). The GDT block is employed to obtain the global channel-wise and spatial-wise dependency. The outputs of GDT are denoted as G_5 and given to a DT module, which is used to encapsulate the dependency from the high-level representatives into the upsampled features. The procedure can be formulated as

$$\begin{aligned} D_4 &= \mathrm{UP}\left(\mathrm{Concat}\left(F_4, \mathrm{DT}_4\left(G_5\right)\right)\right), \\ D_i &= \mathrm{UP}\left(\mathrm{Concat}\left(F_i', \mathrm{DT}_i\left(D_{i+1}\right)\right)\right), \quad i = 2, 3, \\ D_1 &= \mathrm{Concat}\left(F_1', \mathrm{DT}_1\left(D_2\right)\right), \end{aligned} \tag{3}$$

where $DT_i(\cdot)$ is the ith DT block, and UP$(\cdot)$ and Concat$(\cdot,\cdot)$ are the upsampling and concatenation operations, respectively. Deep supervision is performed on D_2 and D_3, utilizing the resized masks of HV and PV as references.

3.3. Vessel-growing decoder

The size of the feature map computed by the encoder becomes 1/512 of the input. This downsampling process poses a challenge for vessels with small radii, resulting in a possible missing of the corresponding features. While decoding, the features corresponding to thinner vessels will be restored with upsampling and skip connection. However, if the restored features are not accurately traced to the correct source, errors may propagate to associated downstream vessels. This is primarily because HV and PV are sparsely distributed in space and exhibit minimal differences in morphological, and intensity-based semantic information but significant differences in their spatial positional semantics and topological structure semantics. To reduce misclassifications of minor vessels, two points need to be addressed.

1. Minimizing the impact of redundant information while preserving effective semantic information at the channel level. Thus directing the model's attention to channels that are crucial for segmenting targets.
2. During the decoding process, make the model focus on the relative positions of HV and PV, and concentrate on the target region at the spatial level

To achieve the above points, we constructed a top-down topology-aware decoder, consisting mainly of two components: GDT (Global Dual Transformer) and DT (Dependency Transmitting) blocks, for channel and spatial dependency capturing.

The GDT block applies channel and spatial self-attention mechanisms. The channel self-attention mechanism removes redundant semantic information, while the spatial self-attention mechanism establishes global spatial dependencies, thereby focusing on the regions of HV and PV. Meanwhile, DT module employs a dynamic weighting structure that adaptively performs both channel and spatial attention simultaneously. Compared to sequential/parallel channel and spatial attention, this approach allows each pixel in the spatial domain to adaptively select the semantic information to focus on. Last but not least, the deep supervision mechanism, employed during decoding, encourages the model to concentrate on the target regions at the spatial level, effectively removing spatial redundancy. We now illustrate the detailed structure of the GDT and DT blocks in the following section.

3.3.1. Global dual transformer (GDT)

The Global Dual Transformer module is positioned at the end of the encoder, applying the self-attention mechanism to capture global spatial and channel-wise dependencies in high-level features, as shown in Fig. 3. The process begins with a Swin-transformer (Liu et al., 2021) block to extract window-based local pixel-wise dependencies from F_4, which comprise a regular window-based multi-head self-attention and a shifted window-based multi-head self-attention. The obtained features are downsampled through patch merging, denoted as F_g. Subsequently, the channel-wise self-attention block obtains each channel's Query, Key, and Value through grouped convolutions, where the number of groups is equal to the number of channels. Compared to regular convolutions, group convolutions can generate Q, K and V for each channel independently, which is inaccessible by regular convolution. By isolating computations for each channel, group convolutions enhance the model's ability to process and emphasize distinct channel characteristics, enabling more focused attention on critical channels through self-attention mechanisms. Thus, features on each channel will be projected into three cubes to perform self-attention. This process is expressed as

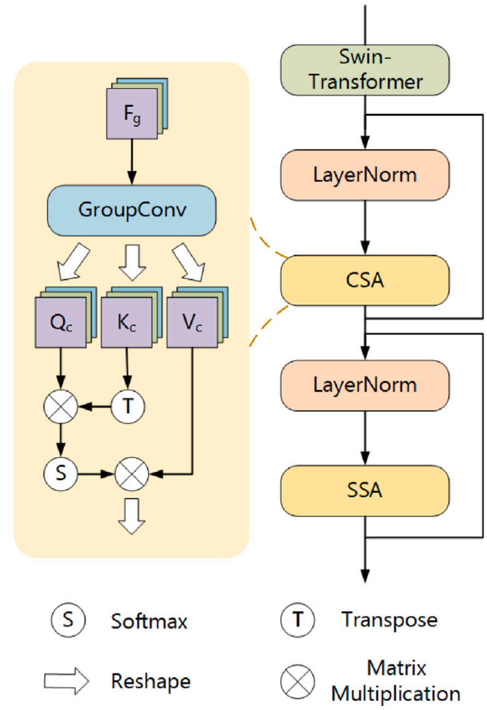

Fig. 3. The structure of global dual transformer (GDT module).

$$\left\{Q_c, K_c, V_c\right\}_j = \mathrm{GConv}(F_g), \quad j = 1, \ldots, C \tag{4}$$

where GConv is the projection function with j as channels' index, and Q_c, K_c, and V_c are the projected representations of Q, K, and V, respectively. Afterward, the output is reshaped into a $3C \times W\,H\,D$ matrix to facilitate channel-wise self-attention. Lastly, the channel-dependent features undergo a pixel-wise multi-head self-attention unit to capture global dependencies in the spatial domain. The whole process can be written as

$$\begin{aligned} G_{tmp} &= \mathrm{CSA}\left(\mathrm{LN}\left(F_g\right)\right) \oplus F_g, \\ G_5 &= \mathrm{SSA}\left(\mathrm{LN}\left(G_{tmp}\right)\right) \oplus G_{tmp}, \end{aligned}$$

where CSA$(\cdot)$ and SSA$(\cdot)$ are channel-wise self-attention and spatial-wise self-attention, and LN$(\cdot)$ denotes layer normalization, respectively. The attention is calculated as

$$Attention\left(Q, K, V\right) = \mathrm{Softmax}\left(\frac{QK^{\top}}{\sqrt{d}}\right)V, \tag{5}$$

with Q, K the d-dimensional feature vectors.

3.3.2. Dependency transmitting block

The dual convolutional structure is introduced to map dependencies into the upsampled image, drawing inspiration from Zhao et al. (2023). The detailed process is depicted in Fig. 4. Specifically, grouped convolution is utilized to compute the upsampled high-level features, with the number of convolution groups matching the number of channels. The

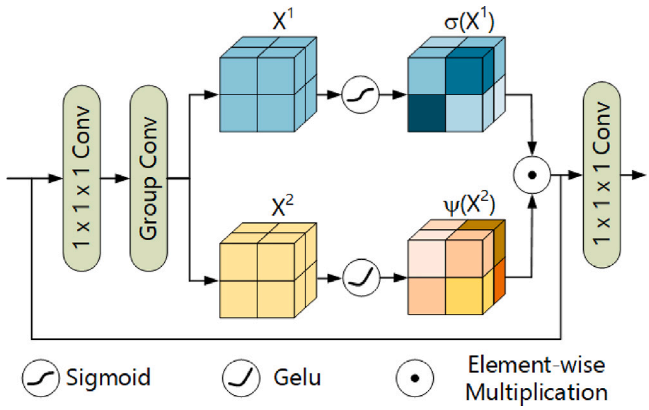

Fig. 4. The structure of dependency transmitting block (DT block).

characteristics of each channel are calculated twice so that the output channels' number is twice the input features. The output features are split channel-wise into two equal parts. Each part comprises a feature cube computed from the preceding channel cube, with positions corresponding to the two groups. Sigmoid and GELU functions are employed to activate the two groups, and the results are multiplied. Finally, a convolution with the kernel size of $3 \times 3 \times 3$ is employed to reach the final output result. This entire process can be expressed as

$$\begin{aligned} X_j^1, X_j^2 &= \mathrm{GConv}\left(\mathrm{Conv}_1\left(Y_j\right)\right), \qquad j = 1, \dots, C \\ \tilde{Y}_j &= \sigma\left(X_j^1\right) \otimes \psi\left(X_j^2\right), \qquad j = 1, \dots, C \\ Y^{out} &= \mathrm{Conv}_1\left(\tilde{Y} \oplus Y\right), \end{aligned} \tag{6}$$

where $\tilde{Y}_j$ is the cube of jth channel on original features, X_j^1, X_j^2 are the grouped features of $\tilde{Y}_j$, and ψ is the GELU activation. The calculated features and encoder features are concatenated, and then the output of the decoder layer is obtained by a convolution operation.

3.4. Multi-task learning mechanism

The proposed structure assembles three output heads to process the obtained feature from the decoder. Specifically, the obtained features D_1 in Function (3) are fed to three parallel residual blocks with fully connected layers and activation functions to obtain corresponding representatives to the three tasks, i.e.,

$$\left\{D_{vs}, D_{cl}, D_{es}\right\} = \mathcal{R}(D_1), \tag{7}$$

with $\mathcal{R}(\cdot)$ the set of three residual blocks. Here, D_{vs}, D_{cl}, and D_{es} represent the outputs from different heads for vessel segmentation, centerline regression, and edge segmentation, respectively. The main branch produces pixel-level prediction maps for HV and PV segmentation, while the other two auxiliary heads produce the predicted vessel centerline regression map and vessel contour segmentation map. The ancillary tasks assist the segmentation head by capturing the topological structure and improving the connectivity of peripheral HV and PV. In implementation, vessel and contour segmentation branches utilize the softmax function for activation, whereas the activation function of the centerline regression branch employs sigmoid. The main task and the two corresponding auxiliary tasks are as below:

1. Main task (HV/PV Segmentation): The main task is to segment HV and PV from a CT volume, that is supervised by their labeled masks.

2. Auxiliary task 1 (Centerline Regression): To characterize the skeletons of HV and PV with the same importance for minor and main vessels, the centerline regression is employed. We use a distance map as the reference for the regression task. Existing distance maps are unable to describe the topological structure of liver vessels because some make the peripheral vessels less salient (Xue et al., 2020; Ma et al., 2020; Liu et al., 2022), and others require additional labeled landmarks (Zhang et al., 2021a). We propose a distance map based on vessel centerlines to emphasize the tree structure of vessels, especially in focusing on the path from peripheral vessels to the trunk. Specifically, the values of pixels within the vessel area are their spatial distances from the vessel centerline, and a mapping function is used to rescale the values in $(0, 1]$. Values of all pixels outside the vessel area are set as 0. This distance map can be defined as:

$$M_d(x) = \begin{cases} 0, & x \in \mathcal{R}_{out} \\ \mathcal{F}\left(\min_{y \in C} \|x - y\|^2\right), & x \in \mathcal{R}_{in} \cup \mathcal{S} \end{cases} \tag{8}$$

where $\mathcal{R}_{in}$ and $\mathcal{R}_{out}$ denote the inside and outside the region of the target, $\mathcal{S}$ designates the surface of the target, C is the centerline of veins, and $\mathcal{F}(\cdot)$ is the mapping function, which is defined as:

$$\mathcal{F}(d) = \frac{1}{\log(e + d)}, \tag{9}$$

where d is the calculated distance. Compared with SDM, the proposed heatmap makes peripheral vessels more salient and emphasizes the trace between the trunk and small vessels.

3. Auxiliary task 2 (Edge Segmentation): The edge segmentation is exploited to preserve connectivities between small and superior vessels. Due to the difference in pixel value intensity between the segmented target and the background, there are usually observable gradient changes near the contour of the segmented target. Detecting those gradient changes helps to obtain a complete segmentation boundary, thereby ensuring the connectivity of the segmentation target. The binary contour map indicates the mask of segmentation boundaries, which is used to prompt the model to compute the segmentation boundaries of the target area. However, because the average pixel difference between veins and liver parenchyma is slight, it leads to ambiguity in labeling peripheral vessels. Hence, there will be bias in labeling vessel contours, making it hard to mark all areas with significant gradient changes around the target. In addition, when using the contour map to optimize the model, there will be a severe imbalance between positive pixels and backgrounds because of the slenderness of boundaries. To extract the area of gradient change as much as possible and alleviate the inequality of positive and negative pixels, we propose an expanded contour map. We denote the labeled HV and PV Masks as M_o and perform dilation and erosion operations on M_o, respectively. The obtained masks are designated as M_d and M_e. The edge of M_o, M_d, and M_e are denoted as C_o, C_d, and C_e, correspondingly. Then, the expanded contour map C_{exp} is

$$C_{exp} := C_o \cup C_d \cup C_e. \tag{10}$$

The proposed expanded contour map is employed as the reference for the edge segmentation task.

3.5. Loss function

The loss function is defined as the summation of loss for HV/PV segmentation (L_{se}), centerline regression (L_{cr}), edge segmentation (L_{ec}), and deep supervision (L_{deep}). Moreover, L_{se}, L_{ec}, and L_{deep} are calculated by the addition of Dice Loss and Binary Cross-entropy Loss reads

$$L_{se,\ ec,\ deep} = 1 - \frac{2\sum_{i=1}^{N} y_i \hat{y}_i}{\sum_{i=1}^{N} y_i + \sum_{i=1}^{N} \hat{y}_i} - \frac{1}{N}\sum_{i=1}^{N} y_i \log \hat{y}_i, \tag{11}$$

where y_i and $\hat{y}_i$ are the original label and prediction of a pixel, respectively. L_{cr} is obtained by computing MSE Loss

$$L_{cr} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2. \tag{12}$$

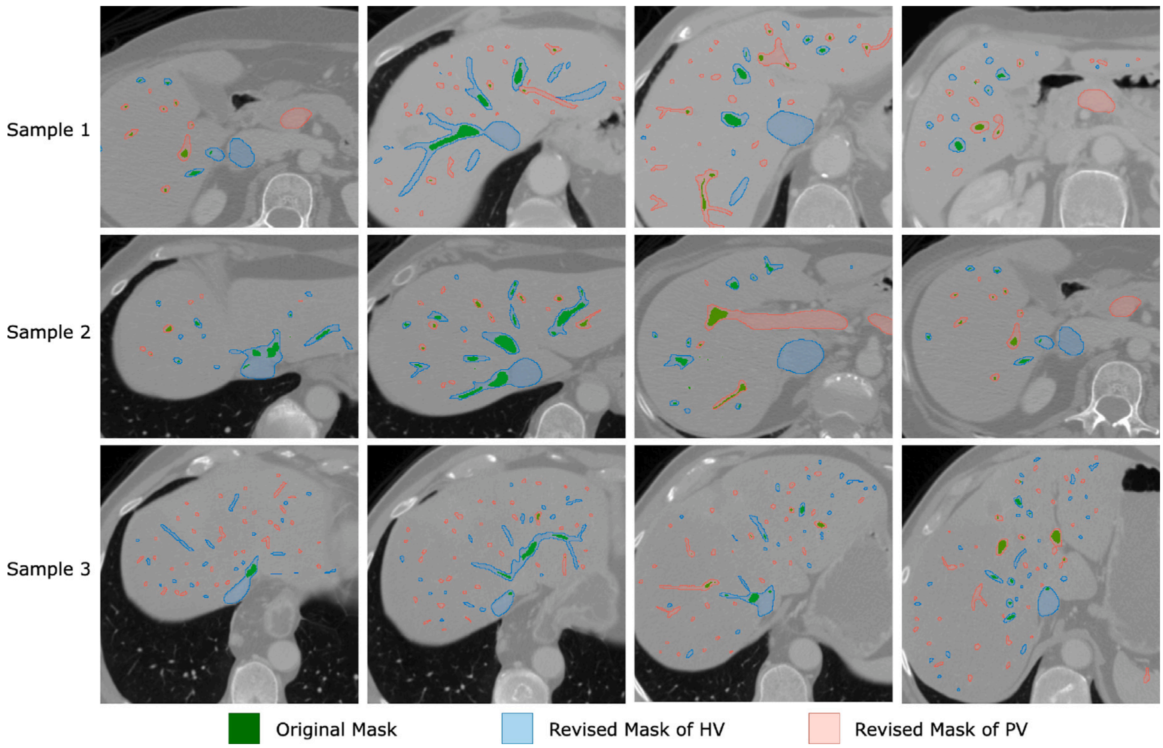

Fig. 5. The examples of slice-level revision. Areas in green are the original masks of hepatic vessels, whereas areas in blue and red are relabeled annotations of HV and PV.

The total loss function is defined as

$$L_{total}(x) = L_{se}(x) + \alpha L_{cr}(x) + \beta L_{ec}(x) + \gamma L_{deep}(x), \tag{13}$$

weighted by hyperparameters α, β and γ.

4. Experiment

4.1. Dataset

Datasets with ground truth are the foundation of deep learning. Popular public datasets for liver vessel segmentation include 3DIRCADb1 (Soler et al., 2010) and MSD (Antonelli et al., 2022). In particular, 3DIRCADb1 contains 20 CT volumes with liver edges, liver vessels, and liver tumors annotated; Task08 (one of the ten sub-datasets for liver vessel segmentation) contains 303 CT volumes with labeled vessel masks and tumors in the training set, and 140 cases in the test set with no annotations. Both datasets have obvious shortcomings, e.g., the size of 3DIRCADb1 is small, and many omissions/faults in the annotations of MSD while the hepatic and portal veins are not annotated separately. To this end, we propose and publish so far the largest dataset for liver vessel segmentation. We re-annotate the original Task08 from MSD to avoid repeated data collection, and distinguish the corresponding masks of hepatic and portal veins. Moreover, the original dataset is divided into two parts with the threshold of layer thickness at 2 mm, denoted as *Thin-slice Dataset* (layer larger than 2 mm) and *Thick-slice Dataset* (layer smaller than 2 mm). As a result, Thin-slice Dataset contains 61 volumes, while Thick-slice Dataset contains 242 volumes. The details of the two datasets can be found in Table 1, where slice thickness and pixel spacing are in millimeters. Given the well-maintained continuity of hepatic vessels in volumes with small thicknesses, we opt to validate the effectiveness of the proposed model with other alternative methods using the Thin-slice Dataset.

Through the re-annotation of the dataset, twelve experts in radiology from Tsinghua ChangGung Hospital and the First Affiliated Hospital of Xi'an Jiaotong University participated. Every image was annotated independently by two radiological experts in a slice-by-slice manner. The inconsistent parts of the annotations were handed over to a third expert for judgment. The final decision was produced by majority voting among three experts. The entire annotation processing for one case in the Thin-slice Dataset took roughly 2 h, while it took about 45 min in the Thick-slice Dataset. Statistics are shown in Table 2. 10.52% of original pixel-level annotations are revised, and 80.24% of relabeled pixels are newly added. In Fig. 5, we display 12 slices from three examples to illustrate how the annotations are revised. The figure shows that the new annotations cover many areas of main and minor vessels that were originally missed. Some comparisons of 3D structure for original annotations and relabeled masks are shown in Fig. 6.

Table 1
Traits of proposed thin-slice dataset and thick-slice dataset.

Dataset	Statistics	Range	Mean	Medium
Thin	Slice thickness	$[0.8, 2]$	1.49	1.5
	Number of slice	$[102, 181]$	147.3	148.0
	Pixel spacing	$[0.623, 0.977]$	0.790	0.785
Thick	Slice thickness	$[2.5, 8]$	4.934	5.0
	Number of slice	$[24, 177]$	50.1	45.0
	Pixel spacing	$[0.568, 0.977]$	0.809	0.805

Table 2
Comparisons between original and revised data. The mean and medium are calculated at volume level.

	Count	Percentile	Mean (%)	Medium (%)
Revised	0.606 M	10.52	11.60	8.97
Increased	2.092 M	80.24	79.54	81.45

4.2. Implementation details

In preprocessing, four patches with a size of $96 \times 96 \times 96$ are randomly cropped from the original volume. Data augmentation methods, such as random flips on three axes, random rotation, and random intensity shift, are adopted to enrich the training data for overfitting

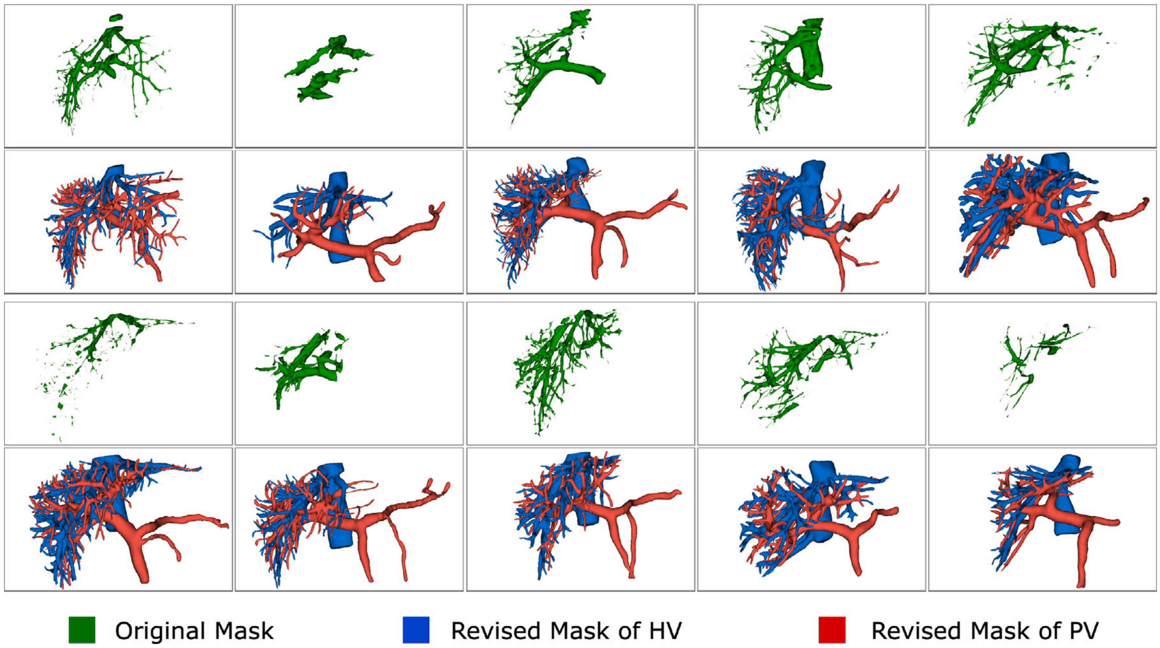

Fig. 6. Comparison between original labels and revised labels. 10 out of 303 revisions are shown. The first and third rows depict the original liver vessel masks in the MSD dataset, while the corresponding positions in the second and fourth rows showcase the modified masks with separate labels for hepatic vein (HV) and portal vein (PV).

Table 3
Ablation studies on the proposed method (HV and PV represent the Dice of HV and PV).

	Baseline	Multi-task learning	GD Transformer	DT Block	HV	PV
Model 1	✓				0.789	0.773
Model 2	✓	✓			**0.811**	**0.798**
Model 3	✓		✓		0.807	0.792
Model 4	✓			✓	0.788	0.778
Model 5	✓	✓	✓		0.816	**0.802**
Model 6	✓		✓	✓	**0.817**	0.800
Model 7	✓	✓		✓	0.812	0.796
Model 8	✓	✓	✓	✓	**0.826**	**0.807**

elimination. Model training utilizes the AdamW (Loshchilov and Hutter, 2017) optimizer with initial learning rate of 10^{-3} and weight decay set at 10^{-5}. The batch size is 16, comprising 4 patches from each of the four volumes. The training stops when there is no increase in the Dice score on the validation set within 2000 epochs. The training hyperparameters and stopping criteria remain consistent across both the proposed method and the comparison models. The experiments are implemented using the PyTorch framework and conducted on 8 NVIDIA A100 GPUs with 40 GB memory.

4.3. Evaluation metrics

Three segmentation evaluation metrics are considered for evaluating the performance of our model, i.e., the *dice coefficient* (Dice) for overall segmentation and per category, *Voxel Segmentation Precision*, Sensitivity, and 95% Hausdorff distance (95HD) that reads

$$95\text{HD}(Y, \hat{Y}) = \max\left\{ q_{95}(d_{Y\hat{Y}}),\ q_{95}(d_{\hat{Y}Y}) \right\} \tag{14}$$

with $d_{Y\hat{Y}}$ and $d_{\hat{Y}Y}$ defined as

$$d_{Y\hat{Y}} = \left\{ \min_{\hat{y}\in\hat{Y}}\ d(y,\hat{y}), y \in Y \right\},\ d_{\hat{Y}Y} = \left\{ \min_{y\in Y}\ d(\hat{y},y), \hat{y} \in \hat{Y} \right\}, \tag{15}$$

where $d(\cdot,\cdot)$ is the distance between two points, and $q_{95}()$ is the function calculating 95% quantile.

The pairwise t-tests are employed to examine whether there are significant differences between different methods. Specifically, the mean Dice score of predicted masks in a fold of cross-validation over the entire dataset is considered a data point, and average Dice scores calculated within the same fold are paired in the test.

4.4. Ablation study

Extensive experiments are carried out to evaluate the necessity of the proposed modules, whereby the effectiveness of Global Dual Transformer, Dependency Transmitting blocks, and multi-task learning are validated. Moreover, we explore the efficiency of different branches in multi-task learning and the usefulness of dual self-attention in GDT. The Thin-slice dataset is randomly divided into a ratio of 4 : 1 for training and test sets.

4.4.1. Effectiveness of proposed modules

The UNet is set as the baseline in the experiment. We first apply each proposed module individually to observe the impact of each module on the Dice of HV and PV. Then, we assess the mutual promotion between modules by employing every combination of two modules. Lastly, we integrate all the modules to showcase the overall performance of the proposed model. Experimental results are reported in Table 3.

For Models 1–4, in comparison to the baseline model, multi-task learning and the Global Dual Transformer (GDT) demonstrate increases in the Dice of HV by 0.022 and 0.018, and for PV by 0.025 and 0.019, respectively. Conversely, the Dual Transformer (DT) Block shows a minor influence on improving the segmentation of HV and PV. Intuitively, without the computation of dual dependencies on high-level features,

Table 4
Ablation studies on multi-task learning with different backbones. (HV and PV represent the Dice of HV and PV.).

Backbones	Multi-task	HV	PV
UNet++		0.804	0.770
	✓	0.815	0.786
LVSNet		0.801	0.781
	✓	0.814	0.790
Swin-UNETR		0.808	0.793
	✓	0.821	0.795
nnFormer		0.812	0.795
	✓	0.816	0.792

Table 5
Ablation studies on auxiliary tasks.

Auxiliary task	Whole vessel	Hepatic vein	Portal vein
Baseline	0.781	0.789	0.773
Center	0.797	0.802	0.791
Edge	0.792	0.799	0.784
Center+edge	**0.805**	**0.811**	**0.798**

Table 6
Ablation studies on GDT.

	Whole vessel	Hepatic vein	Portal vein
Baseline	0.783	0.788	0.778
Swin	0.796	0.799	0.792
Swin+CSA	0.805	0.811	0.799
Swin+SSA	0.802	0.806	0.798
Swin+CSA+SSA	**0.809**	**0.817**	**0.800**

DT blocks have limited functionality. When two modules are combined, as seen in Model 6, it can be inferred that the model achieves the highest Dice in the segmentation of HV among Models 5–7, utilizing the synergy of GDT and DT Blocks, forming the complete Vessel-growing Decoder. Furthermore, in comparison to the base model, the use of the vessel-growing decoder increases the Dice for HV and PV by 0.028 and 0.027, respectively. From Model 3 and Model 6, with the assistance of DT blocks, the performance of GDT improves by 0.010 and 0.005 on the Dice for HV and PV, indicating that the transmission of dependencies enhances the model's efficiency. Referring to Model 2–3 and Model 5, we find that after multi-task learning and GDT are assembled, the performance exceeds that of either of them singly adopted. Thus, these two modules can promote each other. Finally, when adopting all the proposed modules as Model 8, the highest Dice of HV and PV from Model 1–7 are promoted by 0.009 and 0.005, respectively.

As shown in Table 3, multi-task learning significantly improved the Dice scores for HV and PV segmentation by 0.022 and 0.025, respectively. To further evaluate the effectiveness of this module, we integrated it with different backbones to assess its impact across various models. We compared four backbones: UNet++, LVSNet, Swin-UNETR, and nnFormer, with the corresponding results reported in Table 4. Integrating the multi-task learning module with the UNet++, LVSNet, and Swin-UNETR backbones resulted in an improvement of over 0.01 in the Dice scores for HV segmentation. The Dice scores for PV segmentation with the assembled UNet++ and LVSNet also improved by approximately 0.01. However, for the nnFormer, the multi-task learning module did not significantly enhance the segmentation metrics. This may be due to the large number of parameters in nnFormer, which can extract and retain central line-related topological information and gradient changes related to vessel boundaries during the decoding process, thus achieving similar results to those obtained by introducing the multi-task learning module. Nevertheless, given the large number of parameters in nnFormer, the multi-task learning module presents a more cost-effective solution.

Table 7
Ablation studies on the Q, K, and V generation in CSA of GDT.

	Whole vessel	Hepatic vein	Portal vein
Regular Conv.	0.802	0.811	0.792
Group Conv.	**0.809**	**0.817**	**0.800**

4.4.2. *Effectiveness of auxiliary tasks*

For validation of proposed branches in multi-task learning, we examine the impact of using different tasks on segmentation efficiency. To observe the impact of each task distinctly, we use UNet as the baseline model and results are depicted in Table 5. Here, *center* and *edge* represent centerline regression branch and edge segmentation branch, respectively. Specifically, using the centerline regression or edge segmentation task independently to assist the main task can enhance the Dice of HV and PV by at least 0.01. When the two auxiliary tasks are adopted simultaneously, the segmentation performance of the model will be further improved.

4.4.3. *Effectiveness of global dual transformer*

The impact of CSA and SSA in the GDT is also verified. We use UNet + DT blocks as the baseline model (Model 4 in Table 3). We compare the following scenarios: baseline, only using Swin-transformer, Swin-transformer combined with either CSA or SSA, and Swin-transformer assembled with both CSA and SSA (GDT). The results are shown in Table 6 with *Swin* representing the Swin-transformer. We can deduce from the table that the Swin-transformer is capable of extracting long-distance spatial dependence, thus improving the model's performance. Because CSA can remove redundant information by the self-attention operations between channels, while SSA can extract global dependencies, both can further improve the model's efficiency. When the two are employed simultaneously, the model achieves a better effect.

We evaluated different methods for extracting the channel's Q, K, and V components within the CSA of the GDT module. Specifically, we compared the impact of using group convolutions, which extract Q, K, and V for each channel independently, with regular convolutions that fuse channel information to generate these components. The results presented in Table 7 show that using group convolutions leads to an improvement in the Dice coefficient for the hepatic vein and portal vein by 0.006 and 0.008, respectively.

4.5. *Hyperparameters in loss function*

We compared the impact of different values for three hyperparameters (α, β, and λ) in the loss function on the average Dice. Each hyperparameter is assigned values of 0.01, 0.1, and 1, and all combinations of these values were evaluated. The result are detailed in Fig. 7. We observe that regardless of the value of the deep supervision loss hyperparameter λ, the average Dice score increases as the weight of the centerline regression loss α or the boundary segmentation loss β increases. When α and β are set to 0.01, the Dice score shows a noticeable increase with higher values of λ, while when α and β are set to 0.1 or 1, the Dice score becomes less sensitive to increases in the deep supervision weight, and may even slightly decline. Similarly, as the value of λ increases, the contribution of higher values of α and β to the Dice score enhancement diminishes. The comparison indicates that multi-task learning and deep supervision may share overlapping mechanisms in improving the Dice score. Based on these results, we ultimately set the values of α, β, and λ to 1, 1, and 0.1, respectively.

4.6. *Method comparison*

We conduct a comprehensive comparison of our method with other popular models, including UNet (Ronneberger et al., 2015), VNet (Milletari et al., 2016), UNet++ (Zhou et al., 2019), LVSNet (Yan et al., 2020), nnUNet (Isensee et al., 2021), TransUNet (Chen et al., 2023),

Table 8
Methods' comparison on the proposed Thin-slice Dataset.

Method	Dice	Precision	Sensitivity	95HD (mm)	*p*-value	Dice per class	
						HV	PV
UNet (Ronneberger et al., 2015)	0.786 ± 0.014	0.803 ± 0.019	0.776 ± 0.018	22.35 ± 5.36	< 0.001	0.808 ± 0.018	0.764 ± 0.014
VNet (Milletari et al., 2016)	0.778 ± 0.017	0.796 ± 0.022	0.763 ± 0.018	26.33 ± 8.51	< 0.001	0.789 ± 0.019	0.767 ± 0.018
UNet++ (Zhou et al., 2019)	0.793 ± 0.014	0.812 ± 0.019	0.785 ± 0.020	20.85 ± 7.29	< 0.001	0.813 ± 0.016	0.772 ± 0.022
LVSNet (Yan et al., 2020)	0.799 ± 0.015	0.814 ± 0.016	0.785 ± 0.018	23.08 ± 6.64	0.002	0.808 ± 0.019	0.789 ± 0.022
nnUnet (Isensee et al., 2021)	0.802 ± 0.017	0.819 ± 0.015	0.793 ± 0.018	19.71 ± 5.89	0.002	0.811 ± 0.014	0.793 ± 0.018
TransUNet (Chen et al., 2023)	0.783 ± 0.014	0.796 ± 0.019	0.775 ± 0.016	23.66 ± 8.32	< 0.001	0.795 ± 0.016	0.770 ± 0.019
UNETR (Hatamizadeh et al., 2022b)	0.766 ± 0.018	0.752 ± 0.020	0.778 ± 0.018	25.79 ± 7.48	< 0.001	0.778 ± 0.017	0.754 ± 0.019
Swin-UNETR (Hatamizadeh et al., 2022a)	0.801 ± 0.015	0.824 ± 0.013	0.792 ± 0.016	19.11 ± 5.25	< 0.001	0.816 ± 0.016	0.786 ± 0.020
nnFormer (Zhou et al., 2023)	0.795 ± 0.019	0.807 ± 0.018	0.786 ± 0.021	21.03 ± 6.72	< 0.001	0.812 ± 0.021	0.787 ± 0.018
CoTUNet (Wu et al., 2023)	0.772 ± 0.018	0.779 ± 0.019	0.762 ± 0.023	23.42 ± 6.28	< 0.001	0.784 ± 0.016	0.759 ± 0.022
OriNet (Jiang et al., 2024)	0.768 ± 0.015	0.777 ± 0.019	0.761 ± 0.013	25.61 ± 4.26	< 0.001	0.775 ± 0.016	0.760 ± 0.012
VSNet (Ours)	**0.816 ± 0.019**	**0.835 ± 0.017**	**0.810 ± 0.014**	**17.35 ± 6.06**		**0.824 ± 0.024**	**0.807 ± 0.018**

Table 9
Methods' comparison on the 3DIRCADb..

Method	Dice	Precision	Sensitivity	95HD(mm)	*p*-value	Dice per class	
						HV	PV
UNet (Ronneberger et al., 2015)	0.883 ± 0.025	0.923 ± 0.024	0.890 ± 0.027	20.59 ± 5.35	0.002	0.894 ± 0.018	0.872 ± 0.022
VNet (Milletari et al., 2016)	0.861 ± 0.018	0.931 ± 0.022	0.865 ± 0.019	23.15 ± 4.81	< 0.001	0.869 ± 0.021	0.853 ± 0.018
UNet++ (Zhou et al., 2019)	0.897 ± 0.027	0.913 ± 0.024	0.891 ± 0.028	20.32 ± 5.33	< 0.001	0.906 ± 0.026	0.888 ± 0.023
LVSNet (Yan et al., 2020)	0.907 ± 0.018	0.928 ± 0.016	0.889 ± 0.019	17.36 ± 4.68	0.096	0.910 ± 0.021	0.903 ± 0.023
nnUnet (Isensee et al., 2021)	0.906 ± 0.017	0.921 ± 0.013	0.891 ± 0.017	18.58 ± 3.08	0.077	0.914 ± 0.012	0.898 ± 0.018
TransUNet (Chen et al., 2023)	0.885 ± 0.017	0.898 ± 0.021	0.872 ± 0.018	20.06 ± 5.45	< 0.001	0.891 ± 0.019	0.878 ± 0.020
UNETR (Hatamizadeh et al., 2022b)	0.868 ± 0.015	0.859 ± 0.016	0.875 ± 0.018	26.22 ± 3.79	< 0.001	0.876 ± 0.017	0.862 ± 0.018
Swin-UNETR (Hatamizadeh et al., 2022a)	0.910 ± 0.019	0.921 ± 0.021	0.896 ± 0.026	16.22 ± 4.41	0.138	**0.923 ± 0.025**	0.896 ± 0.024
nnFormer (Zhou et al., 2023)	0.908 ± 0.023	0.917 ± 0.025	0.902 ± 0.018	16.53 ± 5.81	< 0.001	0.916 ± 0.025	0.899 ± 0.022
CoTUNet (Wu et al., 2023)	0.871 ± 0.016	0.887 ± 0.011	0.856 ± 0.018	19.72 ± 5.06	< 0.001	0.881 ± 0.017	0.860 ± 0.020
OriNet (Jiang et al., 2024)	0.859 ± 0.011	0.874 ± 0.019	0.844 ± 0.013	21.39 ± 5.59	< 0.001	0.875 ± 0.016	0.843 ± 0.019
VSNet (Ours)	**0.915 ± 0.023**	**0.937 ± 0.021**	**0.902 ± 0.018**	**15.98 ± 5.06**		**0.922 ± 0.016**	**0.907 ± 0.024**

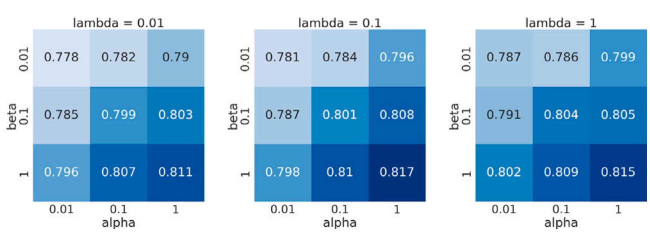

Fig. 7. The average dice with different hyperparameter setting.

Table 10
Numbers of parameters of each model. M stands for million.

Method	Parameters
UNet (Ronneberger et al., 2015)	17.8M
VNet (Milletari et al., 2016)	45.6M
UNet++ (Zhou et al., 2019)	59.4M
LVSNet (Yan et al., 2020)	34.7M
nnUNet (Isensee et al., 2021)	22.4M
TransUNet (Chen et al., 2021)	38.5M
UNETR (Hatamizadeh et al., 2022b)	92.8M
Swin-UNETR (Hatamizadeh et al., 2022a)	62.2M
nnFormer (Zhou et al., 2023)	149.9M
CoTUNet (Wu et al., 2023)	33.3M
OriNet (Jiang et al., 2024)	50.8M
VSNet (Ours)	**38.1M**

UNETR (Hatamizadeh et al., 2022b), Swin-UNETR (Hatamizadeh et al., 2022a), and nnFormer (Zhou et al., 2023). Moreover, method (Jiang et al., 2024) developed for cardiovascular parts and another one (Wu et al., 2023) for pulmonary vessels are compared. The numbers of each model's parameter can be found in Table 10. All approaches are trained on the same dataset, using identical training hyperparameters and stopping criteria as the proposed method. We employ a 5-fold cross-validation for both the proposed model and alternative approaches for evaluation.

4.6.1. Thin-slice dataset

The numeric results are shown in Table 8. The average Dice of a fold over the whole dataset is treated as a data point for calculating standard deviation. From the table, we can observe that the proposed model achieves the highest scores in all criteria. For the overall Dice of hepatic vessel segmentation, the proposed method outperforms Swin-UNETR at 0.015, which reaches the highest rank for hepatic vessel segmentation on the MSD challenge leaderboard[2] without utilizing external data for training. For HV and PV, the average Dice of the proposed model are 0.008 and 0.018 higher than Swin-UNETR and LVSNet, respectively.

4.6.2. 3DIRCADb1 dataset

Experiments are conducted on 3DIRCADb1 dataset to evaluate the generalization of each method. The quantitative results are reported in Table 9, from which we can observe that all the metrics are significantly improved for all the methods. The reason for the progress could be the omission of small vessels in the dataset, which reduces the difficulties in segmenting minor vessels and thereby leads to higher segmentation performance (Yan et al., 2020; Keshwani et al., 2020). Under this scenario, the proposed method does not present apparent advantages, since the ability to segment minor vessels of the proposed model is not fully utilized.

4.6.3. Visual comparison

The examples of predicted masks generated by different methods are illustrated in Fig. 8, with corresponding selected zoomed areas

[2] https://decathlon-10.grand-challenge.org/evaluation/challenge/leaderboard/

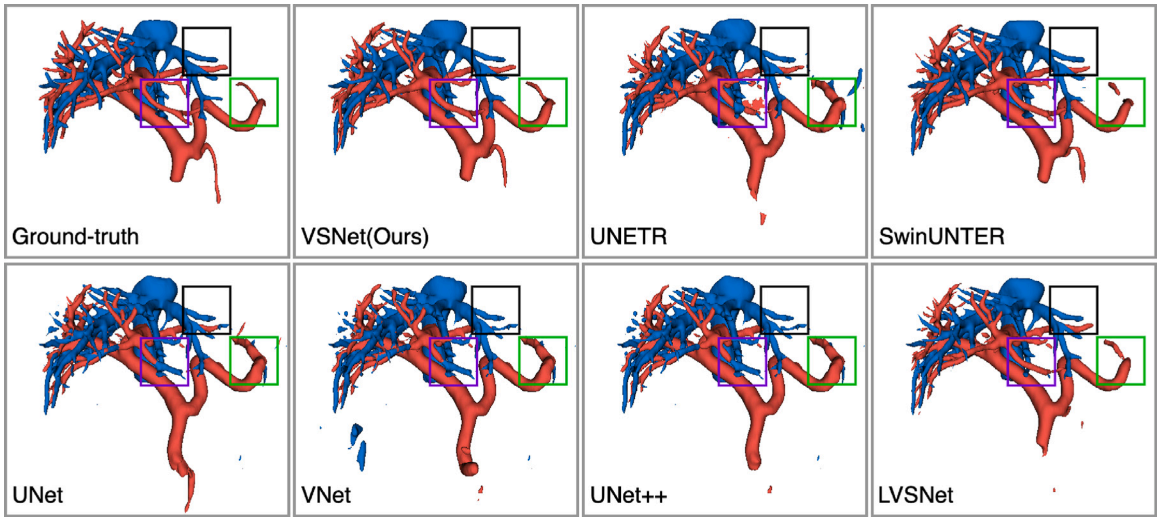

Fig. 8. Visual results of the ground truth, the proposed methods, and the alternative approaches. HV and PV are stained in blue and red. We selected three regions containing minor vessels on each image, marked in green, purple, and black respectively. The zoomed patches can be found in Fig. 9.

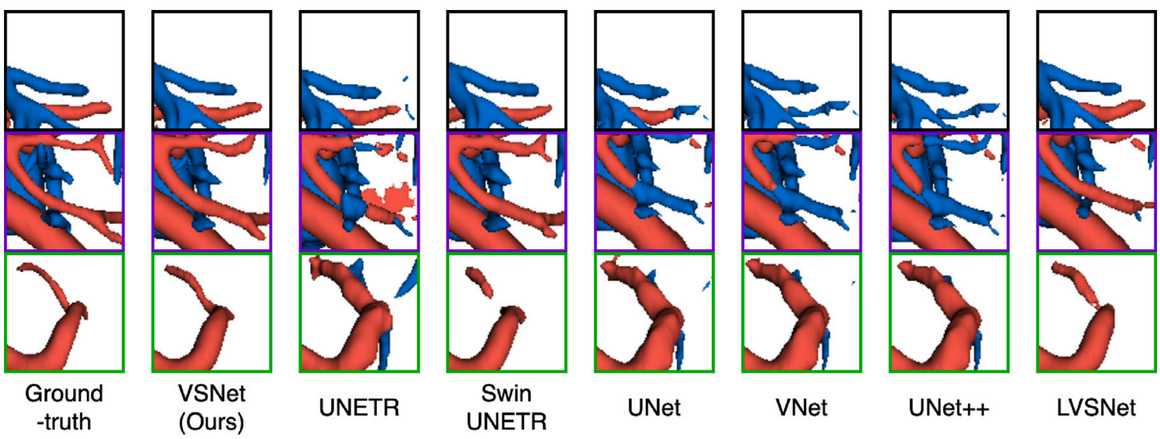

Fig. 9. Zoomed patches of minor vessel.

shown in Fig. 9 to emphasize the segmentation results of minor vessels. All methods successfully segment the trunks of HV and PV. However, for peripheral vessels, existing methods exhibit some notable drawbacks. Most CNN-based methods, such as UNet and LVSNet, mistakenly classify close minor vessels into opposite categories, as highlighted by the misclassifications in the patches with purple squares. Conversely, transformer-based methods produce some fragments. In contrast, the proposed method better alleviates these issues. Yet, it still leads to omissions of small vessels.

In Fig. 8, the part of the hepatic vein in the purple box is misclassified as portal vein in the UNet predictions compared to the ground truth. This vessel is a major branch of the middle hepatic vein, and located in segment IV of the liver. However, when it is misclassified, the adjacent liver parenchyma is incorrectly assigned to segment III. Since preoperative planning for surgical incisions is often based on segment boundaries, such segmentation errors can lead to incorrect designs of the surgical resection plane.

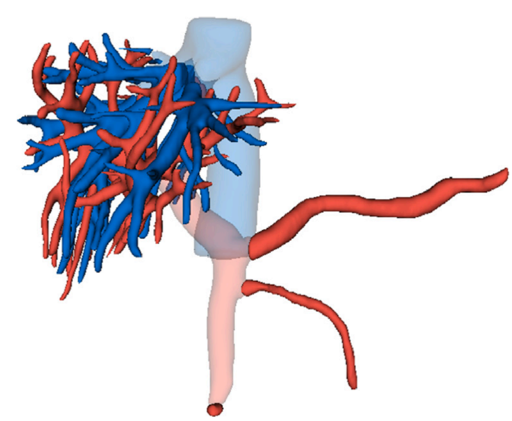

Fig. 10. Comparison of complete hepatic vessels and minor vessels.

4.7. Evaluation of minor vessels

As illustrated in the previous subsection, most existing models can accurately segment the trunk of liver vessels, which cover most pixels of the entire vessel. However, in clinical practice, doctors need to analyze the tendencies of minor hepatic vessels for designing surgical plans. Therefore, calculating the overall Dice cannot exhibit the model efficiency of segmenting small vessels (Yan et al., 2020). To evaluate the model's efficiency in segmenting minor vessels, we use the masks of HVs and HVs without corresponding main vessels as the references. To avoid additional manual annotations of main vessels, referring to Zheng et al. (2020), we develop an automatic approach to remove the trunks based on the erosion–dilation mechanism. First, the erosion operation

Table 11
Methods' comparison on minor vessels.

Method	Dice per class	
	HV	PV
UNet	0.564 ± 0.028	0.543 ± 0.026
VNet	0.550 ± 0.022	0.526 ± 0.025
UNet++	0.573 ± 0.028	0.552 ± 0.029
LVSNet	0.617 ± 0.023	0.595 ± 0.021
UNETR	0.533 ± 0.029	0.521 ± 0.026
Swin-UNETR	0.596 ± 0.028	0.578 ± 0.026
VSNet (Ours)	**0.639** ± **0.023**	**0.628** ± **0.024**

is conducted on the original mask for d times. Second, two connected domains, which cover most pixels, are selected as the eroded truck of HV and PV. Then, the eroded trunks are dilated for $d + 2$ times to obtain the coarsen mask of the trunks. An additional two times of dilation is conducted to cover more pixels on the trunks. Finally, we delete the intersection of the original mask and coarsen trunks from the original mask to acquire the masks of minor vessels. An example of extracted minor vessels and original vessels is shown in Fig. 10. We conduct comparison experiments on the proposed dataset small vessel segmentation. The results are shown in Table 11. From the table, we can discover that the proposed model is 0.022 and 0.033 higher than the closest results of small vessel segmentation in HV and PV, respectively, while in experiments utilizing the complete mask, the proposed method is only 0.009 and 0.016 ahead, respectively.

4.8. Generalization to other datasets of 3D vessels

We present the performance of VSNet on other publicly available vascular segmentation datasets to validate its capacity for segmenting other vascular structures. Numerous vascular structure segmentation datasets have now been made publicly available. For instance, the ASOCA challenge (Gharleghi et al., 2022, 2023), ImageCas (Zeng et al., 2023) datasets for coronary artery segmentation, PARSE challenge dataset (Luo et al., 2023) for pulmonary artery segmentation, and the Carotid Artery Vessel Wall Segmentation Challenge dataset[3] for carotid artery segmentation.

We adopt ImageCas dataset and PARSE challenge dataset to evaluate the generalization ability. The ImageCas dataset contains 1,000 cases with slice thicknesses ranging from 0.25 mm to 0.45 mm. We randomly selected 200 cases to conduct experiments. The training set of PARSE challenge includes 100 annotated CT volumes of the pulmonary artery, with a uniform slice thickness of 1 mm. All cases from the PARSE challenge were included in our experiments. Similar to the HV/PV segmentation experiments, we utilize five-fold cross-validation to assess the effectiveness of the model.

For ImageCas dataset, we compared the proposed method with six segmentation methods: UNet, nnUNet, TransUNet, Swin-UNet, nnFormer, and OriNet which is a model specially developed for cardiovascular parts segmentation. The result reported in Table 12 indicates that VSNet outperforms all competing methods in terms of Dice, including the OriNet which is specifically designed for cardiovascular parts segmentation. and widely recognized high-performance general methods such as nnUNet and Swin-Transformer. All p-values for the hypothesis tests conducted on the Dice scores are less than 0.05, revealing proposed method significantly surpasses other methods in coronary artery segmentation. Lastly, our method also achieved the best results in recall, precision, and 95 HD metrics.

For the PARSE2022 dataset, we compared the proposed method with the same five general methods and an algorithm specifically developed for pulmonary artery segmentation. The results, reported in Table 13, show the proposed method outperformed all comparative methods in terms of Dice. At a significance level of 0.05, VSNet outperformed all methods except nnUNet; at a significance level of 0.1, VSNet also outperformed nnUNet. Additionally, the proposed method achieved the best results in precision and 95 HD. As demonstrated in Section 4.7, the proposed method exhibits a notable improvement in segmenting peripheral vessels. Compared to the ImageCas dataset, the PARSE dataset contains targets with thicker vessel trunks, resulting in less pronounced performance improvement by the proposed method on the PARSE dataset compared to the ImageCas dataset.

5. Discussion

Our proposed method achieves superior results on metrics such as Dice and the 95% Hausdorff distance. However, accurately segmenting hepatic veins (HV) and portal veins (PV) remains challenging due to their complex and variable anatomical structures, low contrast between vessels and the background in CT images, and the indistinct vascular boundaries. By correcting labels in the MSD Task08 dataset and separately annotating the HV and PV labels, we have published the largest dataset for HV and PV segmentation to date. Notably, some original cases in the dataset may have large slice intervals, resulting in missing vessel details and even discontinuities. While these cases may be redundant for model training intended for clinical deployment, we have corrected the labels to enhance the dataset for research purposes.

While the proposed model has achieved state-of-the-art performance, there remains room for improvement. Specifically, when capturing topological structures, the method enhances the model's perception of topology through auxiliary tasks. However, it does not directly incorporate prior knowledge of tubular structures into its framework. Future work could focus on integrating more direct tubular structure detectors to extract vascular structures, such as Sobel operator (Ghadiri et al., 2011), which is sensitive to tube edges. Additionally, exploring deformable-convolution-based methods could be beneficial, as they enable filters to dynamically adjust sampling locations on tubular regions with learnable offsets. Post-processing techniques may also be considered to further improve output accuracy, enhancing the system's practical applicability in clinical applications.

We conducted experiments to explore the hyperparameters of the loss function (see Section 4.5 and Fig. 7) by testing all combinations of 10 hyperparameters. Although hyperparameter search is computationally expensive, we plan to conduct additional experiments to find their global optima for improved performance. Additionally, other hyperparameters, such as patch size, may be examined. Regarding the model, we tested the proposed module on different backbones, all of which showed improvements. We will also explore backbones that integrate language and vision models, such as Universal Model (Liu et al., 2023) and MedSAM (Ma et al., 2024), to further investigate the potential of deep learning-based vascular segmentation algorithms.

Lastly, the MSD dataset that has been refined and re-annotated is the largest and most comprehensive dataset. However, given that it has been published for over six years, its slice thickness and intensity stability may not compare favorably to those of private datasets used in more recent studies. To address this issue, we plan to annotate and release a larger and higher-quality hepatic vessel dataset, which will also include detailed liver segment labels.

6. Conclusion

We propose a novel deep neural network for hepatic and portal vein segmentation by employing two additional output heads for centerline regression and edge segmentation. Besides, two modules (GD Transformer, and DT blocks) are also proposed to improve the efficiency in segmenting the hepatic vein and portal vein. Furthermore, we revise and publish the labeled masks of training data in the MSD dataset. The proposed method outperforms state-of-the-art methods on the revised

[3] https://vessel-wall-segmentation.grand-challenge.org/

Table 12
Methods' comparison on ImageCAS Dataset.

Method	Dice	Precision	Sensitivity	95HD (mm)	*p*-value
UNet (Ronneberger et al., 2015)	0.778 ± 0.015	0.785 ± 0.018	0.772 ± 0.011	35.73 ± 4.27	< 0.001
nnUnet (Isensee et al., 2021)	0.804 ± 0.011	0.801 ± 0.013	0.809 ± 0.009	26.13 ± 3.89	< 0.001
TransUNet (Chen et al., 2021)	0.771 ± 0.017	0.766 ± 0.019	0.779 ± 0.015	32.65 ± 4.11	< 0.001
Swin-UNETR (Hatamizadeh et al., 2022a)	0.796 ± 0.014	0.787 ± 0.017	0.803 ± 0.013	25.33 ± 5.03	0.002
nnFormer (Zhou et al., 2023)	0.787 ± 0.015	0.775 ± 0.013	0.791 ± 0.016	26.67 ± 3.74	< 0.001
OriNet (Jiang et al., 2024)	0.789 ± 0.014	0.783 ± 0.011	0.795 ± 0.013	35.69 ± 4.85	< 0.001
VSNet (Ours)	**0.823 ± 0.016**	**0.829 ± 0.019**	**0.814 ± ± 0.018**	**24.41 ± 4.92**	

Table 13
Methods' comparison on PARSE Dataset.

Method	Dice	Precision	Sensitivity	95HD (mm)	*p*-value
UNet (Ronneberger et al., 2015)	0.834 ± 0.015	0.827 ± 0.017	0.845 ± 0.018	8.93 ± 3.68	< 0.001
nnUNet (Isensee et al., 2021)	0.867 ± 0.013	0.858 ± 0.017	**0.875 ± 0.014**	5.52 ± 1.68	0.061
TransUNet (Chen et al., 2021)	0.829 ± 0. 019	0.834 ± 0.017	0.818 ± 0.023	8.23 ± 3.36	< 0.001
Swin-UNETR (Hatamizadeh et al., 2022a)	0.857 ± 0.014	0.866 ± 0.018	0.851 ± 0.017	6.15 ± 2.89	0.010
nnFormer (Zhou et al., 2023)	0.861 ± 0.013	0.866 ± 0.017	0.853 ± 0.012	6.34 ± 2.79	0.030
CoTUNet (Wu et al., 2023)	0.843 ± 0.014	0.837 ± 0.019	0.846 ± 0.013	9.22 ± 4.51	< 0.001
VSNet (Ours)	**0.874** ± **0.014**	**0.879** ± **0.016**	**0.867** ± **0.011**	**5.14** ± **1.74**	

dataset with a series of ablation studies based on the baseline model conducted on the proposed dataset.

While the proposed method has achieved SOTA results in HV/PV segmentation, compared with tasks such as multi-organ segmentation and tumor segmentation, there is room for improvement in terms of Dice scores, especially compared with multi-organ and tumor segmentation. This observation motivated us to release a revised (also, the largest) dataset, hoping to encourage more researchers' participation in the community.

CRediT authorship contribution statement

Jichen Xu: Writing – original draft, Software, Methodology, Investigation, Formal analysis. **Anqi Dong:** Writing – review & editing, Validation, Supervision, Conceptualization. **Yang Yang:** Writing – review & editing, Validation. **Shuo Jin:** Data curation. **Jianping Zeng:** Data curation. **Zhengqing Xu:** Data curation. **Wei Jiang:** Validation. **Liang Zhang:** Validation, Supervision. **Jiahong Dong:** Supervision, Data curation. **Bo Wang:** Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is partially supported by Natural Science Foundation of China (62088102), Hubei Province Key Research and Development Program, China (2023BCB151), Wuhan Municipal Natural Science Foundation Exploration Program (Chenguang Program), China (2024040801020202), Qin Chuangyuan Cited the High-Level Innovation and Entrepreneurship Talent Project of China (QCYRCXM-2022-367), Tsinghua University Initiative Scientific Research Program of Precision Medicine, China (2022ZLA004), Shaanxi Province Postdoctoral Research Project Funding, China (2023BSHYDZZ56), and Natural Science Basic Research Program of Shaanxi, China (2023-JC-YB-559).

Data availability

The data used in the manuscript are published on the article's Github page.

References

Antonelli, M., Reinke, A., Bakas, S., Farahani, K., Kopp-Schneider, A., Landman, B.A., Litjens, G., Menze, B., Ronneberger, O., Summers, R.M., et al., 2022. The medical segmentation decathlon. Nat. Commun. 13, 4128.

Asrani, S.K., Devarbhavi, H., Eaton, J., Kamath, P.S., 2019. Burden of liver diseases in the world. J. Hepatol. 70, 151–171.

Chelaramani, S., Gupta, M., Agarwal, V., Gupta, P., Habash, R., 2021. Multi-task knowledge distillation for eye disease prediction. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. pp. 3983–3993.

Chen, S., Bortsova, G., García-Uceda Juárez, A., Van Tulder, G., De Bruijne, M., 2019. Multi-task attention-based semi-supervised learning for medical image segmentation. In: Medical Image Computing and Computer Assisted Intervention–MICCAI 2019: 22nd International Conference, Shenzhen, China, October 13–17, 2019, Proceedings, Part III 22. Springer, pp. 457–465.

Chen, J., Lu, Y., Yu, Q., Luo, X., Adeli, E., Wang, Y., Lu, L., Yuille, A.L., Zhou, Y., 2021. TransUNet: Transformers make strong encoders for medical image segmentation. arXiv preprint arXiv:2102.04306.

Chen, J., Mei, J., Li, X., Lu, Y., Yu, Q., Wei, Q., Luo, X., Xie, Y., Adeli, E., Wang, Y., et al., 2023. 3D TransUNet: Advancing medical image segmentation through vision transformers. arXiv preprint arXiv:2310.07781.

Chen, Z., Xie, H., Hu, M., Huang, T., Hu, Y., Sang, N., Zhao, Y., 2020a. Recent progress in treatment of hepatocellular carcinoma. Am. J. Cancer Res. 10, 2993.

Chen, Z., Zhu, L., Wan, L., Wang, S., Feng, W., Heng, P.-A., 2020b. A multi-task mean teacher for semi-supervised shadow detection. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 5611–5620.

Dong, C., Xu, S., Dai, D., Zhang, Y., Zhang, C., Li, Z., 2023. A novel multi-attention, multi-scale 3D deep network for coronary artery segmentation. Med. Image Anal. 85, 102745.

Estienne, T., Vakalopoulou, M., Christodoulidis, S., Battistela, E., Lerousseau, M., Carre, A., Klausner, G., Sun, R., Robert, C., Mougiakakou, S., et al., 2019. U-ReSNet: Ultimate coupling of registration and segmentation with deep nets. In: Medical Image Computing and Computer Assisted Intervention–MICCAI 2019: 22nd International Conference, Shenzhen, China, October 13–17, 2019, Proceedings, Part III 22. Springer, pp. 310–319.

Gao, Z., Zong, Q., Wang, Y., Yan, Y., Wang, Y., Zhu, N., Zhang, J., Wang, Y., Zhao, L., 2023. Laplacian salience-gated feature pyramid network for accurate liver vessel segmentation. IEEE Trans. Med. Imaging.

Ghadiri, F., Akbarzadeh-T, M.-R., Haddadan, S., 2011. Vessel segmentation based on sobel operator and fuzzy reasoning. In: 2011 1st International EConference on Computer and Knowledge Engineering. ICCKE, IEEE, pp. 189–194.

Gharleghi, R., Adikari, D., Ellenberger, K., Ooi, S.-Y., Ellis, C., Chen, C.-M., Gao, R., He, Y., Hussain, R., Lee, C.-Y., et al., 2022. Automated segmentation of normal and diseased coronary arteries—The ASOCA challenge. Comput. Med. Imaging Graph. 97, 102049.

Gharleghi, R., Adikari, D., Ellenberger, K., Webster, M., Ellis, C., Sowmya, A., Ooi, S., Beier, S., 2023. Annotated computed tomography coronary angiogram images and associated data of normal and diseased arteries. Sci. Data 10, 128.

Goceri, E., 2016. Automatic labeling of portal and hepatic veins from MR images prior to liver transplantation. Int. J. Comput. Assist. Radiol. Surg. 11, 2153–2161.

van Gulik, T.M., de Graaf, W., Dinant, S., Busch, O.R.C., Gouma, D.J., 2007. Vascular occlusion techniques during liver resection. Dig. Surg. 24, 274–281.

Guo, X., Huang, S., Fu, X., Wang, B., Huang, X., 2015. Vascular segmentation in hepatic CT images using adaptive threshold fuzzy connectedness method. Biomed. Eng. Online 14, 1–11.

Hao, X., Bruce, C., Pislaru, C., Greenleaf, J.F., 2000. A novel region growing method for segmenting ultrasound images. In: 2000 IEEE Ultrasonics Symposium. Proceedings. An International Symposium (Cat. No. 00CH37121), vol. 2. IEEE, pp. 1717–1720.

Hao, W., Zhang, J., Su, J., Song, Y., Liu, Z., Liu, Y., Qiu, C., Han, K., 2022. HPM-Net: Hierarchical progressive multiscale network for liver vessel segmentation in CT images. Comput. Methods Programs Biomed. 224, 107003.

Hatamizadeh, A., Nath, V., Tang, Y., Yang, D., Roth, H.R., Xu, D., 2022a. Swin UNETR: Swin transformers for semantic segmentation of brain tumors in MRI images. In: Brainlesion: Glioma, Multiple Sclerosis, Stroke and Traumatic Brain Injuries: 7th International Workshop, BrainLes 2021, Held in Conjunction with MICCAI 2021, Virtual Event, September 27, 2021, Revised Selected Papers, Part I. Springer, pp. 272–284.

Hatamizadeh, A., Tang, Y., Nath, V., Yang, D., Myronenko, A., Landman, B., Roth, H.R., Xu, D., 2022b. UNETR: Transformers for 3D medical image segmentation. In: Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision. pp. 574–584.

Heriot, A.G., Karanjia, N.D., 2002. A review of techniques for liver resection. Ann. R. Coll. Surg. Engl. 84, 371.

Huang, Q., Sun, J., Ding, H., Wang, X., Wang, G., 2018. Robust liver vessel extraction using 3D U-Net with variant dice loss function. Comput. Biol. Med. 101, 153–162.

Isensee, F., Jaeger, P.F., Kohl, S.A.A., Petersen, J., Maier-Hein, K.H., 2021. nnU-Net: A self-configuring method for deep learning-based biomedical image segmentation. Nature Methods 18, 203–211.

Isgum, I., Staring, M., Rutten, A., Prokop, M., Viergever, M.A., Van Ginneken, B., 2009. Multi-atlas-based segmentation with local decision fusion—application to cardiac and aortic segmentation in CT scans. IEEE Trans. Med. Imaging 28, 1000–1010.

Ivashchenko, O.V., Zhai, Z., Stoel, B.C., Ruers, T.J.M., 2021. Optimization of hepatic vasculature segmentation from contrast-enhanced MRI, exploring two 3D unet modifications and various loss functions. In: Medical Imaging 2021: Image-Guided Procedures, Robotic Interventions, and Modeling, vol. 11598. SPIE, pp. 125–134.

Jiang, W., Li, Y., Jia, Y., Feng, Y., Yi, Z., Chen, M., Wang, J., 2024. Ori-net: Orientation-guided neural network for automated coronary arteries segmentation. Expert Syst. Appl. 238, 121905.

Kang, X., Zhao, Q., Sharma, K., Shekhar, R., Wood, B.J., Linguraru, M.G., 2014. Automatic labeling of liver veins in CT by probabilistic backward tracing. In: 2014 IEEE 11th International Symposium on Biomedical Imaging. ISBI, IEEE, pp. 1115–1118.

Keshwani, D., Kitamura, Y., Ihara, S., Iizuka, S., Simo-Serra, E., 2020. TopNet: Topology preserving metric learning for vessel tree reconstruction and labelling. In: Medical Image Computing and Computer Assisted Intervention–MICCAI 2020: 23rd International Conference, Lima, Peru, October 4–8, 2020, Proceedings, Part VI 23. Springer, pp. 14–23.

Kitrungrotsakul, T., Han, X.-H., Iwamoto, Y., Foruzan, A.H., Lin, L., Chen, Y.-W., 2017. Robust hepatic vessel segmentation using multi deep convolution network. In: Medical Imaging 2017: Biomedical Applications in Molecular, Structural, and Functional Imaging, vol. 10137. SPIE, pp. 269–274.

Kittler, J., 1983. On the accuracy of the sobel edge detector. Image Vis. Comput. 1, 37–42.

Kuang, H., Menon, B.K., Sohn, S.I.L., Qiu, W., 2021. EIS-Net: Segmenting early infarct and scoring ASPECTS simultaneously on non-contrast CT of patients with acute ischemic stroke. Med. Image Anal. 70, 101984.

Lebre, M.-A., Vacavant, A., Grand-Brochier, M., Rositi, H., Abergel, A., Chabrot, P., Magnin, B., 2019. Automatic segmentation methods for liver and hepatic vessels from CT and MRI volumes, applied to the couinaud scheme. Comput. Biol. Med. 110, 42–51.

Li, Y., Zhang, Y., Liu, J.-Y., Wang, K., Zhang, K., Zhang, G.-S., Liao, X.-F., Yang, G., 2022. Global transformer and dual local attention network via deep-shallow hierarchical feature fusion for retinal vessel segmentation. IEEE Trans. Cybern..

Lian, C., Liu, M., Wang, L., Shen, D., 2021. Multi-task weakly-supervised attention network for dementia status estimation with structural MRI. IEEE Trans. Neural Netw. Learn. Syst. 33, 4056–4068.

Liu, M., Li, F., Yan, H., Wang, K., Ma, Y., Shen, L., Xu, M., Initiative, A.D.N., et al., 2020. A multi-model deep convolutional neural network for automatic hippocampus segmentation and classification in Alzheimer's disease. Neuroimage 208, 116459.

Liu, Z., Lin, Y., Cao, Y., Hu, H., Wei, Y., Zhang, Z., Lin, S., Guo, B., 2021. Swin transformer: Hierarchical vision transformer using shifted windows. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 10012–10022.

Liu, S., Wang, H., Li, Y., Li, X., Cao, G., Cao, W., 2022. AHU-MultiNet: Adaptive loss balancing based on homoscedastic uncertainty in multi-task medical image segmentation network. Comput. Biol. Med. 150, 106157.

Liu, J., Zhang, Y., Chen, J.-N., Xiao, J., Lu, Y., A. Landman, B., Yuan, Y., Yuille, A., Tang, Y., Zhou, Z., 2023. Clip-driven universal model for organ segmentation and tumor detection. In: Proceedings of the IEEE/CVF International Conference on Computer Vision. pp. 21152–21164.

Loshchilov, I., Hutter, F., 2017. Decoupled weight decay regularization. arXiv preprint arXiv:1711.05101.

Luo, G., Wang, K., Liu, J., Li, S., Liang, X., Li, X., Gan, S., Wang, W., Dong, S., Wang, W., et al., 2023. Efficient automatic segmentation for multi-level pulmonary arteries: The PARSE challenge. arXiv preprint arXiv:2304.03708.

Ma, J., He, Y., Li, F., Han, L., You, C., Wang, B., 2024. Segment anything in medical images. Nat. Commun. 15, 654.

Ma, J., Wei, Z., Zhang, Y., Wang, Y., Lv, R., Zhu, C., Gaoxiang, C., Liu, J., Peng, C., Wang, L., et al., 2020. How distance transform maps boost segmentation CNNs: An empirical study. In: Medical Imaging with Deep Learning. PMLR, pp. 479–492.

Milletari, F., Navab, N., Ahmadi, S.-A., 2016. V-net: Fully convolutional neural networks for volumetric medical image segmentation. In: 2016 Fourth International Conference on 3D Vision. 3DV, Ieee, pp. 565–571.

Park, S., Kim, G., Oh, Y., Seo, J.B., Lee, S.M., Kim, J.H., Moon, S., Lim, J.-K., Ye, J.C., 2022. Multi-task vision transformer using low-level chest X-ray feature corpus for COVID-19 diagnosis and severity quantification. Med. Image Anal. 75, 102299.

Qian, Y.-j., 2017. Liver vessel segmentation algorithm based on fuzzy enhance and Hessian matrix. In: 2nd International Conference on Biomedical and Biological Engineering 2017. BBE 2017, Atlantis Press, pp. 540–545.

Ramamonjisoa, M., Lepetit, V., 2019. Sharpnet: Fast and accurate recovery of occluding contours in monocular depth estimation. In: Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops.

Ronneberger, O., Fischer, P., Brox, T., 2015. U-net: Convolutional networks for biomedical image segmentation. In: Medical Image Computing and Computer-Assisted Intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18. Springer, pp. 234–241.

Schindera, S.T., Triller, J., Vock, P., Hoppe, H., 2006. Detection of hepatic portal venous gas: Its clinical impact and outcome. Emerg. Radiol. 12, 164–170.

Soler, L., Hostettler, A., Agnus, V., Charnoz, A., Fasquel, J., Moreau, J., Osswald, A., Bouhadjar, M., Marescaux, J., 2010. 3D Image Reconstruction for Comparison of Algorithm Database: A Patient Specific Anatomical and Medical Image Database. Tech. Rep 1, IRCAD, Strasbourg, France.

Su, J., Liu, Z., Zhang, J., Sheng, V.S., Song, Y., Zhu, Y., Liu, Y., 2021. DV-Net: Accurate liver vessel segmentation via dense connection model with D-BCE loss function. Knowl.-Based Syst. 232, 107471.

Sun, L., Shao, W., Zhu, Q., Wang, M., Li, G., Zhang, D., 2023. Multi-scale multi-hierarchy attention convolutional neural network for fetal brain extraction. Pattern Recognit. 133, 109029.

Svobodova, P., Sethia, K., Strakos, P., Varysova, A., 2022. Automatic hepatic vessels segmentation using RORPO vessel enhancement filter and 3D V-Net with variant dice loss function. Appl. Sci. 13, 548.

Tang, S., Yu, X., Cheang, C.F., Liang, Y., Zhao, P., Yu, H.H., Choi, I.C., 2023. Transformer-based multi-task learning for classification and segmentation of gastrointestinal tract endoscopic images. Comput. Biol. Med. 157, 106723.

Wu, Y., Qi, S., Wang, M., Zhao, S., Pang, H., Xu, J., Bai, L., Ren, H., 2023. Transformer-based 3D U-net for pulmonary vessel segmentation and artery-vein separation from CT images. Med. Biol. Eng. Comput. 61, 2649–2663.

Xu, Z., Lu, D., Wang, Y., Luo, J., Jayender, J., Ma, K., Zheng, Y., Li, X., 2021. Noisy labels are treasure: Mean-teacher-assisted confident learning for hepatic vessel segmentation. In: Medical Image Computing and Computer Assisted Intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part I 24. Springer, pp. 3–13.

Xue, Y., Tang, H., Qiao, Z., Gong, G., Yin, Y., Qian, Z., Huang, C., Fan, W., Huang, X., 2020. Shape-aware organ segmentation by predicting signed distance maps. Proc. AAAI Conf. Artif. Intell. 34, 12565–12572.

Yan, Q., Wang, B., Zhang, W., Luo, C., Xu, W., Xu, Z., Zhang, Y., Shi, Q., Zhang, L., You, Z., 2020. Attention-guided deep neural network with multi-scale feature fusion for liver vessel segmentation. IEEE J. Biomed. Health Inf. 25, 2629–2642.

Yang, Z., Farsiu, S., 2023. Directional connectivity-based segmentation of medical images. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 11525–11535.

Yao, J., Shi, Y., Cao, K., Lu, L., Lu, J., Song, Q., Jin, G., Xiao, J., Hou, Y., Zhang, L., 2021. DeepPrognosis: Preoperative prediction of pancreatic cancer survival and surgical margin via comprehensive understanding of dynamic contrast-enhanced CT imaging and tumor-vascular contact parsing. Med. Image Anal. 73, 102150.

Yu, W., Fang, B., Liu, Y., Gao, M., Zheng, S., Wang, Y., 2019. Liver vessels segmentation based on 3D residual U-NET. In: 2019 IEEE International Conference on Image Processing. ICIP, IEEE, pp. 250–254.

Zeng, A., Wu, C., Lin, G., Xie, W., Hong, J., Huang, M., Zhuang, J., Bi, S., Pan, D., Ullah, N., et al., 2023. ImageCas: A large-scale dataset and benchmark for coronary artery segmentation based on computed tomography angiography images. Comput. Med. Imaging Graph. 109, 102287.

Zeng, Y.-z., Zhao, Y.-q., Tang, P., Liao, M., Liang, Y.-x., Liao, S.-h., Zou, B.-j., 2017. Liver vessel segmentation and identification based on oriented flux symmetry and graph cuts. Comput. Methods Programs Biomed. 150, 31–39.

Zhang, X., Cui, Z., Feng, J., Song, Y., Wu, D., Shen, D., 2021a. CorLab-Net: Anatomical dependency-aware point-cloud learning for automatic labeling of coronary arteries. In: Machine Learning in Medical Imaging: 12th International Workshop, MLMI 2021, Held in Conjunction with MICCAI 2021, Strasbourg, France, September 27, 2021, Proceedings 12. Springer, pp. 576–585.

Zhang, Y., Gao, Y., Zhou, G., He, J., Xia, J., Peng, G., Lou, X., Zhou, S., Tang, H., Chen, Y., 2023. Centerline-supervision multi-task learning network for coronary angiography segmentation. Biomed. Signal Process. Control 82, 104510.

Zhang, Y., Li, H., Du, J., Qin, J., Wang, T., Chen, Y., Liu, B., Gao, W., Ma, G., Lei, B., 2021b. 3D multi-attention guided multi-task learning network for automatic gastric tumor segmentation and lymph node classification. IEEE Trans. Med. Imaging 40, 1618–1631.

Zhang, W., Yang, G., Zhang, N., Xu, L., Wang, X., Zhang, Y., Zhang, H., Del Ser, J., de Albuquerque, V.H.C., 2021c. Multi-task learning with multi-view weighted fusion attention for artery-specific calcification analysis. Inf. Fusion 71, 64–76.

Zhao, H., Gou, Y., Li, B., Peng, D., Lv, J., Peng, X., 2023. Comprehensive and delicate: An efficient transformer for image restoration. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. pp. 14122–14132.

Zhao, Y., Wang, X., Che, T., Bao, G., Li, S., 2022. Multi-task deep learning for medical image computing and analysis: A review. Comput. Biol. Med. 106496.

Zheng, S., Zhang, T., Zhuang, J., Wang, H., Liu, J., 2020. A two-stream meticulous processing network for retinal vessel segmentation. arXiv preprint arXiv:2001.05829.

Zhou, H.-Y., Guo, J., Zhang, Y., Han, X., Yu, L., Wang, L., Yu, Y., 2023. nnFormer: Volumetric medical image segmentation via a 3D transformer. IEEE Trans. Image Process..

Zhou, Z., Siddiquee, M.M.R., Tajbakhsh, N., Liang, J., 2019. UNet++: Redesigning skip connections to exploit multiscale features in image segmentation. IEEE Trans. Med. Imaging 39, 1856–1867.